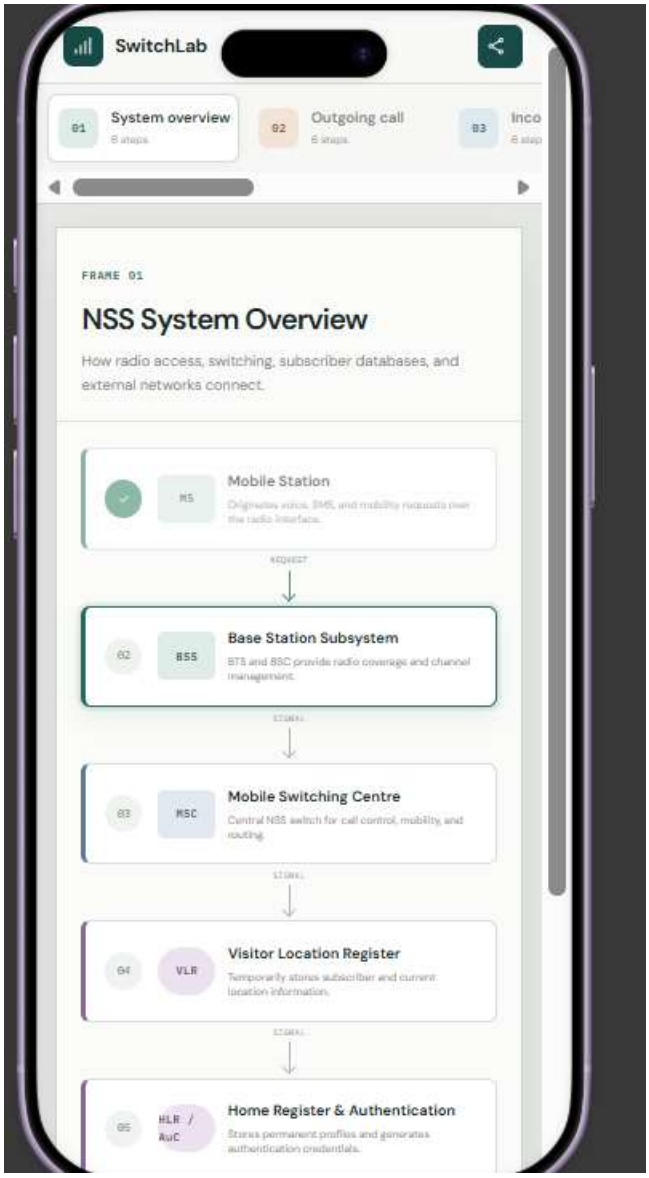
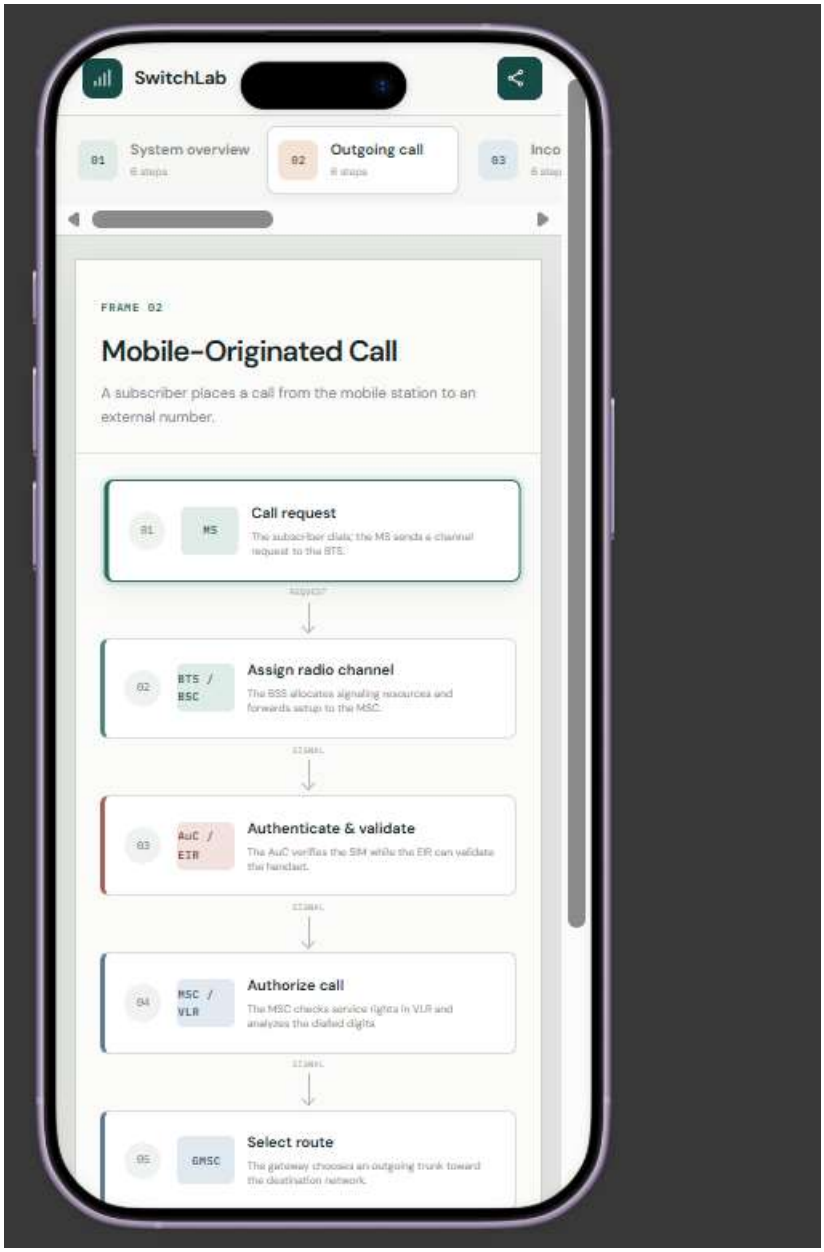
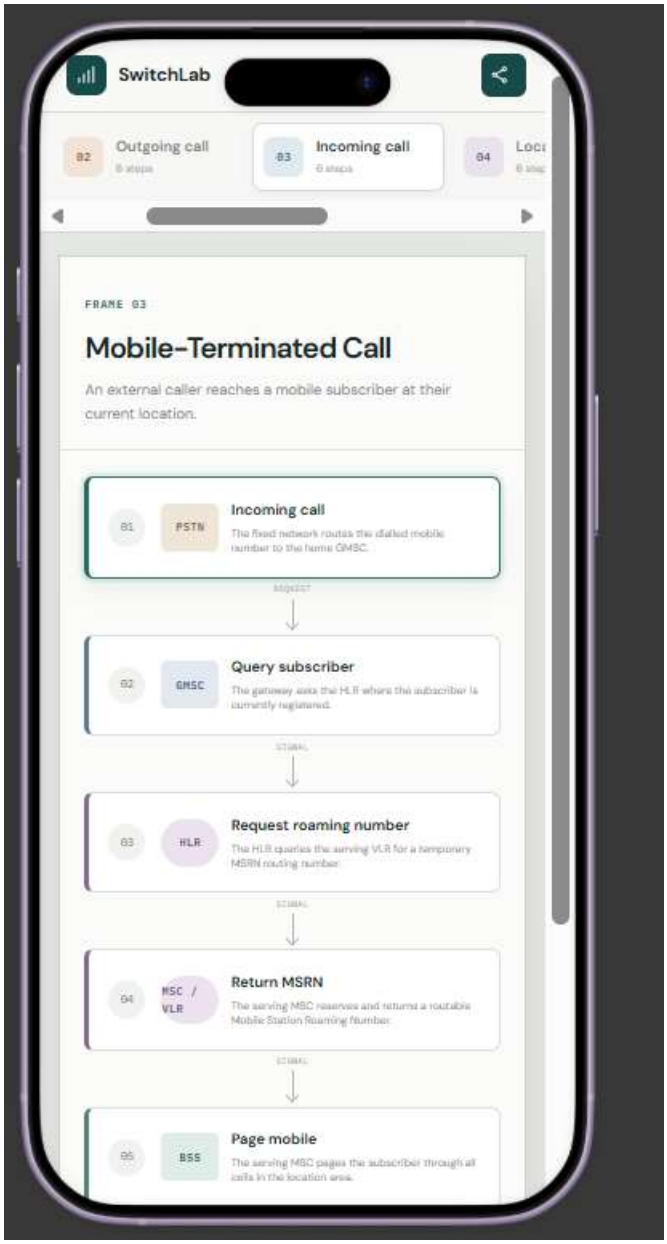
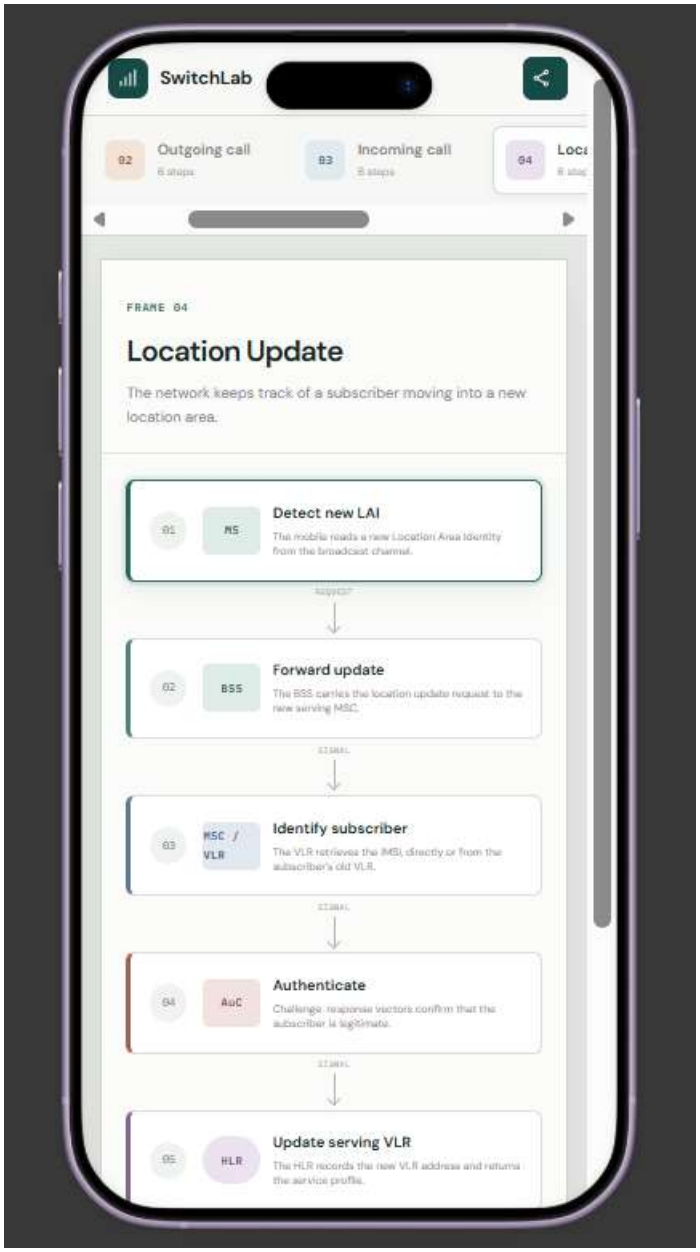
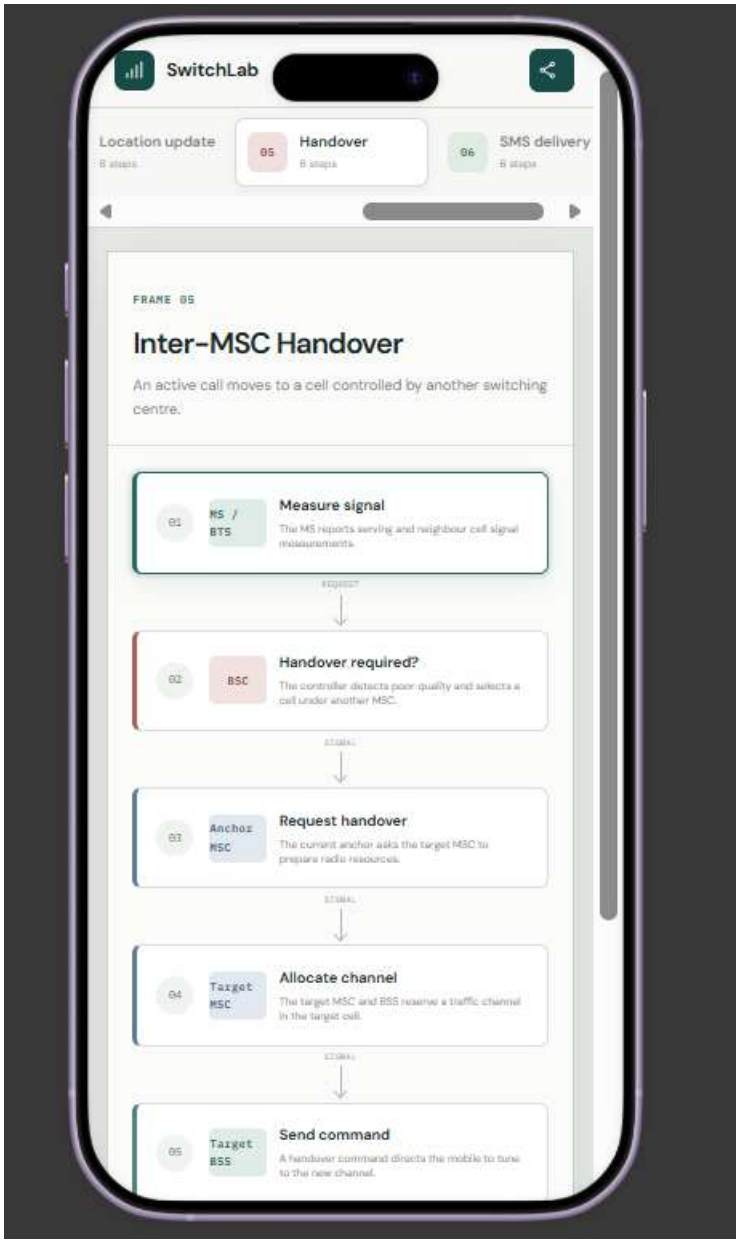
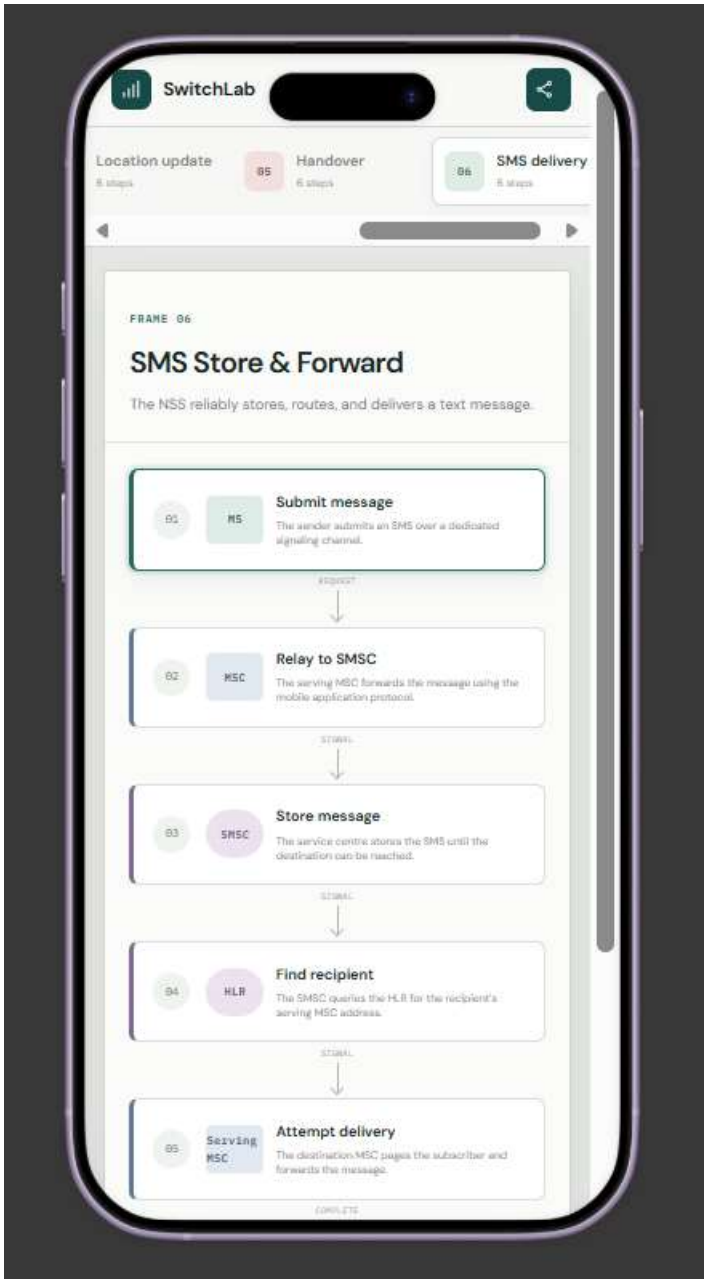

SwitchLab
01
System overview
02
Outgoing call
03
Inco
FRAME 01
NSS System Overview
How radio access, switching, subscriber databases, and external networks connect.
MS
Mobile Station
02
BSS
Base Station Subsystem
BTS and BSC provide radio coverage and channel management.
03
MSC
Mobile Switching Centre
Central NSS switch for call control, mobility, and routing.
04
VLR
Visitor Location Register
Temporarily stores subscriber and current location information.
05
HLR / AuC
Home Register & Authentication
Stores permanent profiles and generates authentication credentials.


SwitchLab
01
System overview
02
Outgoing call
03
FRAME 02
Mobile-Originated Call
A subscriber places a call from the mobile station to an external number.
01
MS
Call request
02
BTS / BSC
Assign radio channel
03
AuC / EIR
Authenticate & validate
04
MSC / VLR
Authorize call
05
GMSC
Select route


SwitchLab
02
Outgoing call
03
Incoming call
04
FRAME 03
Mobile-Terminated Call
An external caller reaches a mobile subscriber at their current location.
01
PSTN
Incoming call
02
GMSC
Query subscriber
03
HLR
Request roaming number
04
MSC / VLR
Return MSRN
05
BSS
Page mobile


SwitchLab
02
Outgoing call
03
Incoming call
04
FRAME 04
Location Update
The network keeps track of a subscriber moving into a new location area.
01
MS
Detect new LAI
02
BSS
Forward update
03
MSC / VLR
Identify subscriber
04
AuC
Authenticate
05
HLR
Update serving VLR


SwitchLab
Location update
05
Handover
06
SMS delivery
FRAME 05
Inter-MSC Handover
An active call moves to a cell controlled by another switching centre.
01
MS / BTS
Measure signal
02
BSC
Handover required?
03
Anchor MSC
Request handover
04
Target MSC
Allocate channel
05
Target BSS
Send command


SwitchLab
Location update
05
Handover
06
SMS delivery
FRAME 06
SMS Store & Forward
The NSS reliably stores, routes, and delivers a text message.
01
MS
Submit message
The sender submits an SMS over a dedicated signaling channel.
02
MSC
Relay to SMSC
The serving MSC forwards the message using the mobile application protocol.
03
SMSC
Store message
The service centre stores the SMS until the destination can be reached.
04
HLR
Find recipient
The SMSC queries the HLR for the recipient's serving MSC address.
05
Serving MSC
Attempt delivery
The destination MSC pages the subscriber and forwards the message.